

Regulatory Affairs

RAHIKA[®] (capmatinib)
150 mg and 200 mg film-coated tablets

Malaysia Package Leaflet (USPI Mar 2023)

IPL Author:	Mohammed Abrar Hussain
CDS Author(s):	Alex Vlantis
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FULL PRESCRIBING INFORMATION

1 INDICATIONS AND USAGE

RAHIKA is indicated for the treatment of adult patients with metastatic non-small cell lung cancer (NSCLC) whose tumors have a mutation that leads to mesenchymal-epithelial transition (MET) exon 14 skipping.

2 DOSAGE AND ADMINISTRATION

2.1 Patient Selection

Select patients for treatment with RAHIKA based on the presence of a mutation that leads to MET exon 14 skipping in tumor or plasma specimens [see *Clinical Studies (14)*]. If a mutation that leads to MET exon 14 skipping is not detected in a plasma specimen, test tumor tissue if feasible.

2.2 Recommended Dosage

The recommended dosage of RAHIKA is 400 mg orally twice daily with or without food.

Swallow RAHIKA tablets whole. Do not break, crush or chew the tablets.

If a patient misses or vomits a dose, instruct the patient not to make up the dose, but to take the next dose at its scheduled time.

2.3 Dosage Modifications for Adverse Reactions

The recommended dose reductions for the management of adverse reactions are listed in Table 1.

Table 1: Recommended RAHIKA Dose Reductions for Adverse Reactions

Dose reduction	Dose and schedule
First	300 mg orally twice daily
Second	200 mg orally twice daily

Permanently discontinue RAHIKA in patients who are unable to tolerate 200 mg orally twice daily.

The recommended dosage modifications of RAHIKA for adverse reactions are provided in Table 2.

Table 2: Recommended RAHIKA Dosage Modifications for Adverse Reactions

Adverse reaction	Severity	Dosage modification
Interstitial Lung Disease (ILD)/Pneumonitis [see <i>Warnings and Precautions (5.1)</i>]	Any grade	Permanently discontinue RAHIKA.
Increased ALT and/or AST without increased total bilirubin [see <i>Warnings and Precautions (5.2)</i>]	Grade 3	Withhold RAHIKA until recovery to baseline ALT/AST. If recovered to baseline within 7 days, then resume RAHIKA at the same dose; otherwise resume RAHIKA at a reduced dose.
	Grade 4	Permanently discontinue RAHIKA.

Adverse reaction	Severity	Dosage modification
Increased ALT and/or AST with increased total bilirubin in the absence of cholestasis or hemolysis <i>[see Warnings and Precautions (5.2)]</i>	ALT and/or AST greater than 3 times ULN with total bilirubin greater than 2 times ULN	Permanently discontinue RAHIKA.
Increased total bilirubin without concurrent increased ALT and/or AST <i>[see Warnings and Precautions (5.2)]</i>	Grade 2	Withhold RAHIKA until recovery to baseline bilirubin. If recovered to baseline within 7 days, then resume RAHIKA at the same dose; otherwise resume RAHIKA at a reduced dose.
	Grade 3	Withhold RAHIKA until recovery to baseline bilirubin. If recovered to baseline within 7 days, then resume RAHIKA at a reduced dose; otherwise permanently discontinue RAHIKA.
	Grade 4	Permanently discontinue RAHIKA.
Increased lipase or amylase <i>[see Warnings and Precautions (5.3)]</i>	Grade 3	Withhold RAHIKA until $\leq$ Grade 2 or baseline. If recovered to baseline or $\leq$ Grade 2 within 14 days, resume RAHIKA at a reduced dose; otherwise permanently discontinue RAHIKA.
	Grade 4	Permanently discontinue RAHIKA.
Pancreatitis <i>[see Warnings and Precautions (5.3)]</i>	Grade 3 or Grade 4	Permanently discontinue RAHIKA.
Hypersensitivity <i>[see Warnings and Precautions (5.4)]</i>	All Grades	If hypersensitivity is suspected based on clinical judgment, withhold RAHIKA until resolution of the event. Permanently discontinue RAHIKA in patients who develop serious hypersensitivity reactions.
Other adverse reactions <i>[see Adverse Reactions (6.1)]</i>	Grade 2	Maintain dose level. If intolerable, consider withholding RAHIKA until resolved, then resume RAHIKA at a reduced dose.
	Grade 3	Withhold RAHIKA until resolved, then resume RAHIKA at a reduced dose.
	Grade 4	Permanently discontinue RAHIKA.

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; ILD, interstitial lung disease; ULN, upper limit of normal.

Grading according to Common Terminology Criteria for Adverse Events (CTCAE) version 5.0.

3 DOSAGE FORMS AND STRENGTHS

Tablets:

- 150 mg: pale orange brown, ovaloid, curved film-coated with beveled edges, unscored, debossed with 'DU' on one side and 'NVR' on the other side
- 200 mg: yellow, ovaloid, curved film-coated with beveled edges, unscored, debossed with 'LO' on one side and 'NVR' on the other side

4 CONTRAINDICATIONS

Hypersensitivity to capmatinib or to any of the excipients [see Description (11)].

5 WARNINGS AND PRECAUTIONS

5.1 Interstitial Lung Disease (ILD)/Pneumonitis

ILD/pneumonitis, which can be fatal, occurred in patients treated with RAHIKA [see *Adverse Reactions* (6.1)].

ILD/pneumonitis occurred in 4.8% of patients treated with RAHIKA in GEOMETRY mono-1, with 1.9% of patients experiencing Grade 3 ILD/pneumonitis and one patient experiencing death (0.3%). Nine patients (2.4%) discontinued RAHIKA due to ILD/pneumonitis. The median time-to-onset of Grade 3 or higher ILD/pneumonitis was 1.8 months (range: 0.2 months to 1.7 years).

Monitor for new or worsening pulmonary symptoms indicative of ILD/pneumonitis (e.g., dyspnea, cough, fever). Immediately withhold RAHIKA in patients with suspected ILD/pneumonitis and permanently discontinue if no other potential causes of ILD/pneumonitis are identified [see *Dosage and Administration* (2.3)].

5.2 Hepatotoxicity

Hepatotoxicity occurred in patients treated with RAHIKA [see *Adverse Reactions* (6.1)]. Increased alanine aminotransferase (ALT)/aspartate aminotransferase (AST) occurred in 15% of patients treated with RAHIKA in GEOMETRY mono-1. Grade 3 or 4 increased ALT/AST occurred in 7% of patients. Three patients (0.8%) discontinued RAHIKA due to increased ALT/AST. The median time-to-onset of Grade 3 or higher increased ALT/AST was 1.8 months (range: 0.5 to 46.4 months).

Monitor liver function tests (including ALT, AST, and total bilirubin) prior to the start of RAHIKA, every 2 weeks during the first 3 months of treatment, then once a month or as clinically indicated, with more frequent testing in patients who develop increased transaminases or bilirubin. Based on the severity of the adverse reaction, withhold, dose reduce, or permanently discontinue RAHIKA [see *Dosage and Administration* (2.3)].

5.3 Pancreatic Toxicity

Elevations in amylase and lipase levels occurred in patients treated with RAHIKA [see *Adverse Reactions* (6.1)]. Increased amylase/lipase occurred in 14% of patients treated with RAHIKA in GEOMETRY mono-1. Grade 3 and 4 increased amylase/lipase occurred in 7% and 1.9% of patients, respectively. Three patients (0.8%) discontinued RAHIKA due to increased amylase/lipase. The median time-to-onset of Grade 3 or higher increased amylase/lipase was 2 months (range: 0.03 to 31.1 months). Pancreatitis (Grade 3) occurred in one patient (0.3%); RAHIKA was permanently discontinued for this event.

Monitor amylase and lipase at baseline and regularly during treatment with RAHIKA. Based on the severity of the adverse reaction, temporarily withhold, dose reduce, or permanently discontinue RAHIKA [see *Dosage and Administration* (2.3)].

5.4 Hypersensitivity Reactions

Serious hypersensitivity reactions occurred in patients treated with RAHIKA in clinical trials other than GEOMETRY mono-1 [see *Adverse Reactions* (6.1)]. Signs and symptoms of hypersensitivity included pyrexia, chills, pruritus, rash, decreased blood pressure, nausea and vomiting. Based on the severity of the adverse reaction, temporarily withhold or permanently discontinue RAHIKA [see *Dosage and Administration* (2.3)].

5.5 Risk of Photosensitivity

Based on findings from animal studies, there is a potential risk of photosensitivity reactions with RAHIKA [see *Nonclinical Toxicology* (13.2)]. In GEOMETRY mono-1, it was recommended that patients use precautionary measures

against ultraviolet exposure such as use of sunscreen or protective clothing during treatment with RAHIKA. Advise patients to limit direct ultraviolet exposure during treatment with RAHIKA.

5.6 Embryo-Fetal Toxicity

Based on findings from animal studies and its mechanism of action, RAHIKA can cause fetal harm when administered to a pregnant woman. Oral administration of capmatinib to pregnant rats and rabbits during the period of organogenesis resulted in malformations at exposures less than the human exposure based on area under the curve (AUC) at the 400 mg twice daily clinical dose. Advise pregnant women of the potential risk to a fetus. Advise females of reproductive potential to use effective contraception during treatment with RAHIKA and for 1 week after the last dose. Advise males with female partners of reproductive potential to use effective contraception during treatment with RAHIKA and for 1 week after the last dose [see *Use in Specific Populations* (8.1, 8.3)].

6 ADVERSE REACTIONS

The following clinically significant adverse reactions are described elsewhere in the labeling:

- ILD/Pneumonitis [see *Warnings and Precautions* (5.1)]
- Hepatotoxicity [see *Warnings and Precautions* (5.2)]
- Pancreatic Toxicity [see *Warnings and Precautions* (5.3)]
- Hypersensitivity reactions [see *Warnings and Precautions* (5.4)]

6.1 Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice.

Metastatic Non-Small Cell Lung Cancer

The safety of RAHIKA was evaluated in GEOMETRY mono-1 [see *Clinical Studies* (14)]. Patients received RAHIKA 400 mg orally twice daily until disease progression or unacceptable toxicity (N = 373). Among patients who received RAHIKA, 37% were exposed for at least 6 months and 22% were exposed for at least one year.

Serious adverse reactions occurred in 53% of patients who received RAHIKA. Serious adverse reactions in $\geq 2\%$ of patients included dyspnea (7%), pneumonia (7%), pleural effusion (4.3%), musculoskeletal pain (3.8%), general physical health deterioration (2.9%), ILD/pneumonitis (2.7%), edema (2.4%), and vomiting (2.4%). Fatal adverse reactions occurred in 0.5% of patients who received RAHIKA, including pneumonitis (0.3%) and death, not otherwise specified (0.3%).

Permanent discontinuation of RAHIKA due to an adverse reaction occurred in 17% of patients. The most frequent adverse reactions ($\geq 1\%$) leading to permanent discontinuation of RAHIKA were ILD/pneumonitis (2.4%), edema (2.4%), fatigue (1.3%), and pneumonia (1.1%).

Dose interruptions due to an adverse reaction occurred in 57% of patients who received RAHIKA. Adverse reactions requiring dosage interruption in $> 2\%$ of patients who received RAHIKA included edema, increased blood creatinine, nausea, increased lipase, vomiting, increased ALT, dyspnea, pneumonia, fatigue, increased amylase, increased AST, musculoskeletal pain, abdominal pain, and increased blood bilirubin.

Dose reductions due to an adverse reaction occurred in 26% of patients who received RAHIKA. Adverse reactions requiring dosage reductions in $> 2\%$ of patients who received RAHIKA included edema, increased ALT and increased blood creatinine.

The most common adverse reactions ($\geq 20\%$) in patients who received RAHIKA were edema, nausea, musculoskeletal pain, fatigue, vomiting, dyspnea, cough, and decreased appetite.

Table 3 summarizes the adverse reactions in GEOMETRY mono-1.

Table 3: Adverse Reactions ($\geq 10\%$) in Patients Who Received RAHIKA in GEOMETRY mono-1

Adverse reactions	RAHIKA (N = 373)	
	Grades 1 to 4 (%)	Grades 3 to 4 (%)
General disorders and administration-site conditions		
Edema ^a	59	13
Musculoskeletal pain ^b	40	4.3
Fatigue ^c	34	8
Pyrexia ^d	14	0.8
Weight decreased	11	0.5
Gastrointestinal disorders		
Nausea	46	2.4
Vomiting	28	2.4
Constipation	19	0.8
Diarrhea	19	0.5
Respiratory, thoracic, and mediastinal disorders		
Dyspnea	25	7
Cough ^e	21	0.5
Pneumonia ^f	13	6
Metabolism and nutrition disorders		
Decreased appetite	21	1.1
Skin and subcutaneous tissue disorders		
Rash ^g	13	0.5
Nervous system disorders		
Dizziness ^h	13	0.5

^aEdema includes edema peripheral, generalized edema, face edema, edema, localized edema, edema genital, eyelid edema, peripheral swelling, scrotal edema, and penile edema.

^bMusculoskeletal pain includes arthralgia, back pain, bone pain, musculoskeletal chest pain, musculoskeletal pain, myalgia, neck pain, non-cardiac chest pain, pain in extremity, pain in jaw, spinal pain.

^cFatigue includes fatigue and asthenia.

^dPyrexia includes pyrexia and body temperature increased.

^eCough includes cough, upper airway cough syndrome, and productive cough.

^fPneumonia includes pneumonia aspiration, pneumonia, pneumonia influenzal, pneumonia bacterial, lower respiratory tract infection, and lung abscess.

^gRash includes rash, dermatitis acneiform, rash maculo-papular, eczema, erythema multiforme, rash macular, dermatitis, rash erythematous, rash pustular, dermatitis bullous, and rash vesicular.

^hDizziness includes dizziness, vertigo, and vertigo positional.

Clinically relevant adverse reactions occurring in $< 10\%$ of patients treated with RAHIKA included pruritus (including allergic pruritus), ILD/pneumonitis, cellulitis, acute kidney injury (including renal failure), urticaria, and acute pancreatitis.

Table 4 summarizes the laboratory abnormalities in GEOMETRY mono-1.

Table 4: Select Laboratory Abnormalities ($\geq 20\%$) Worsening From Baseline in Patients Who Received RAHIKA in GEOMETRY mono-1

Laboratory abnormalities	RAHIKA ^a	
	Grades 1 to 4 (%)	Grades 3 to 4 (%)
Chemistry		
Decreased albumin	72	1.9
Increased creatinine	65	0.5
Increased alanine aminotransferase	39	9
Increased amylase	34	4.7
Increased alkaline phosphatase	32	0.6
Increased gamma-glutamyltransferase	30	6
Increased lipase	29	9
Increased aspartate aminotransferase	28	6
Decreased phosphate	26	4.4
Increased potassium	25	4.1
Decreased sodium	24	6
Decreased glucose	23	0.3
Hematology		
Decreased lymphocytes	45	14
Decreased leukocytes	25	1.7
Decreased hemoglobin	24	2.8

^aThe denominator used to calculate the rate varied from 359 to 364 based on the number of patients with a baseline value and at least one post-treatment value.

Other Clinical Trials Experience

The following adverse reactions have been reported following administration of RAHIKA: hypersensitivity.

7 DRUG INTERACTIONS

7.1 Effect of Other Drugs on RAHIKA

Strong CYP3A Inhibitors

Coadministration of RAHIKA with a strong CYP3A inhibitor increased capmatinib exposure, which may increase the incidence and severity of adverse reactions of RAHIKA [see *Clinical Pharmacology* (12.3)]. Closely monitor patients for adverse reactions during coadministration of RAHIKA with strong CYP3A inhibitors.

Strong and Moderate CYP3A Inducers

Coadministration of RAHIKA with a strong CYP3A inducer decreased capmatinib exposure. Coadministration of RAHIKA with a moderate CYP3A inducer may also decrease capmatinib exposure. Decreases in capmatinib exposure may decrease RAHIKA anti-tumor activity [see *Clinical Pharmacology* (12.3)]. Avoid coadministration of RAHIKA with strong and moderate CYP3A inducers.

7.2 Effect of RAHIKA on Other Drugs

CYP1A2 Substrates

Coadministration of RAHIKA increased the exposure of a CYP1A2 substrate, which may increase the adverse reactions of these substrates [see *Clinical Pharmacology* (12.3)]. If coadministration is unavoidable between RAHIKA and CYP1A2 substrates where minimal concentration changes may lead to serious adverse reactions, decrease the CYP1A2 substrate dosage in accordance with the approved prescribing information.

P-glycoprotein (P-gp) and Breast Cancer Resistance Protein (BCRP) Substrates

Coadministration of RAHIKA increased the exposure of a P-gp substrate and a BCRP substrate, which may increase the adverse reactions of these substrates [see *Clinical Pharmacology* (12.3)]. If coadministration is unavoidable between

RAHIKA and P-gp or BCRP substrates where minimal concentration changes may lead to serious adverse reactions, decrease the P-gp or BCRP substrate dosage in accordance with the approved prescribing information.

MATE1 and MATE2K Substrates

Coadministration of RAHIKA may increase the exposure of MATE1 and MATE2K substrates, which may increase the adverse reactions of these substrates [see *Clinical Pharmacology* (12.3)]. If coadministration is unavoidable between RAHIKA and MATE1 or MATE2K substrates where minimal concentration changes may lead to serious adverse reactions, decrease the MATE1 or MATE2K substrate dosage in accordance with the approved prescribing information.

8 USE IN SPECIFIC POPULATIONS

8.1 Pregnancy

Risk Summary

Based on findings from animal studies and its mechanism of action [see *Clinical Pharmacology* (12.1)], RAHIKA can cause fetal harm when administered to a pregnant woman. There are no available data on RAHIKA use in pregnant women. Oral administration of capmatinib to pregnant rats and rabbits during the period of organogenesis resulted in malformations at maternal exposures less than the human exposure based on AUC at the 400 mg twice daily clinical dose (see *Data*). Advise pregnant women of the potential risk to a fetus.

In the U.S. general population, the estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2% to 4% and 15% to 20%, respectively.

Data

Animal Data

In rats, maternal toxicity (reduced body weight gain and food consumption) occurred at 30 mg/kg/day (approximately 1.4 times the human exposure based on AUC at the 400 mg twice daily clinical dose). Fetal effects included reduced fetal weights, irregular/incomplete ossification, and increased incidences of fetal malformations (e.g., abnormal flexure/inward malrotation of hindpaws/forepaws, thinness of forelimbs, lack of/reduced flexion at the humerus/ulna joints, and narrowed or small tongue) at doses of ≥ 10 mg/kg/day (approximately 0.6 times the human exposure based on AUC at the 400 mg twice daily clinical dose).

In rabbits, no maternal effects were detected at doses up to 60 mg/kg/day (approximately 1.5 times the human exposure based on AUC at the 400 mg twice daily clinical dose). Fetal effects included small lung lobe at ≥ 5 mg/kg/day (approximately 0.016 times the human exposure based on AUC at the 400 mg twice daily clinical dose), and reduced fetal weights, irregular/incomplete ossification and increased incidences of fetal malformations (e.g., abnormal flexure/malrotation of hindpaws/forepaws, thinness of forelimbs/hindlimbs, lack of/reduced flexion at the humerus/ulna joints, small lung lobes, narrowed or small tongue) at the dose of 60 mg/kg/day.

8.2 Lactation

Risk Summary

There are no data on the presence of capmatinib or its metabolites in either human or animal milk or its effects on the breastfed child or on milk production. Because of the potential for serious adverse reactions in breastfed children, advise women not to breastfeed during treatment with RAHIKA and for 1 week after the last dose.

8.3 Females and Males of Reproductive Potential

Based on animal data, RAHIKA can cause malformations at doses less than the human exposure based on AUC at the 400 mg twice daily clinical dose [see *Use in Specific Populations* (8.1)].

Pregnancy Testing

Verify pregnancy status for females of reproductive potential prior to starting treatment with RAHIKA.

Contraception

Females

Advise females of reproductive potential to use effective contraception during treatment with RAHIKA and for 1 week after the last dose.

Males

Advise males with female partners of reproductive potential to use effective contraception during treatment with RAHIKA and for 1 week after the last dose.

8.4 Pediatric Use

Safety and effectiveness of RAHIKA in pediatric patients have not been established.

8.5 Geriatric Use

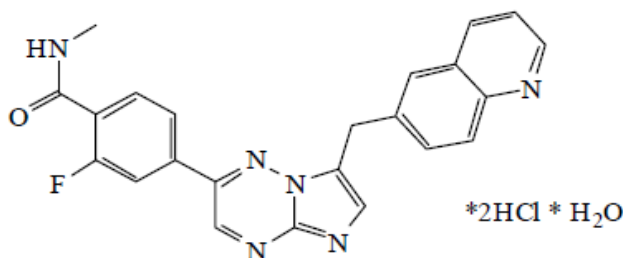
In GEOMETRY mono-1, 61% of the 373 patients were 65 years or older and 18% were 75 years or older. No overall differences in the safety or effectiveness were observed between these patients and younger patients.

8.6 Renal Impairment

No dosage adjustment is recommended in patients with mild (baseline creatinine clearance [CLcr] 60 to 89 mL/min by Cockcroft-Gault) or moderate renal impairment (CLcr 30 to 59 mL/min) [see *Clinical Pharmacology* (12.3)]. RAHIKA has not been studied in patients with severe renal impairment (CLcr 15 to 29 mL/min).

11 DESCRIPTION

Capmatinib is a kinase inhibitor. The chemical name is 2-Fluoro-N-methyl-4-[7-(quinolin-6-ylmethyl)imidazo[1,2-b][1,2,4]triazin-2-yl]benzamide—hydrogen chloride—water (1/2/1). The molecular formula for capmatinib dihydrochloride monohydrate is $C_{23}H_{21}Cl_2FN_6O_2$. The relative molecular mass is 503.36 g/mol for the dihydrochloride monohydrate salt and 412.43 g/mol for the free base. The chemical structure for capmatinib dihydrochloride monohydrate is shown below:



Capmatinib dihydrochloride monohydrate is a yellow powder with a pK_{a1} of 0.9 (calculated) and pK_{a2} of 4.5 (experimentally). Capmatinib dihydrochloride monohydrate is slightly soluble in acidic aqueous solutions at pH 1 and 2 and of further decreasing solubility towards neutral condition. The log of the distribution coefficient (n-octanol/acetate buffer pH 4.0) is 1.2.

RAHIKA is supplied for oral use as ovaloid, curved film-coated tablets with beveled edges, unscored containing 150 mg (pale orange brown color) or 200 mg (yellow color) capmatinib (equivalent to 183.000 mg or 244.000 mg respectively of capmatinib dihydrochloride monohydrate). Each tablet strength contains colloidal silicon dioxide; crospovidone; magnesium stearate; mannitol; microcrystalline cellulose; povidone; and sodium lauryl sulfate as inactive ingredients.

The 150 mg tablet coating contains ferric oxide, red; ferric oxide, yellow; ferrousferic oxide; hypromellose; polyethylene glycol (PEG) 4000; talc; and titanium dioxide. The 200 mg tablet coating contains ferric oxide, yellow; hypromellose; polyethylene glycol (PEG) 4000; talc; and titanium dioxide.

12 CLINICAL PHARMACOLOGY

12.1 Mechanism of Action

Capmatinib is a kinase inhibitor that targets MET, including the mutant variant produced by exon 14 skipping. MET exon 14 skipping results in a protein with a missing regulatory domain that reduces its negative regulation leading to increased downstream MET signaling. Capmatinib inhibited cancer cell growth driven by a mutant MET variant lacking exon 14 at clinically achievable concentrations and demonstrated anti-tumor activity in murine tumor xenograft models derived from human lung tumors with either a mutation leading to MET exon 14 skipping or MET amplification. Capmatinib inhibited the phosphorylation of MET triggered by binding of hepatocyte growth factor or by MET amplification, as well as MET-

mediated phosphorylation of downstream signaling proteins and proliferation and survival of MET-dependent cancer cells.

12.2 Pharmacodynamics

Exposure-Response

Capmatinib exposure-response relationships and the time course of pharmacodynamics response are unknown.

Cardiac Electrophysiology

No large mean increase in QTc (i.e. > 20 ms) was detected following treatment with RAHIKA at the recommended dosage of 400 mg orally twice daily.

12.3 Pharmacokinetics

Capmatinib exposure (AUC_{0-12h} and C_{max}) increased approximately proportionally over a dose range of 200 mg (0.5 times the recommended dosage) to 400 mg. Capmatinib reached steady-state by day 3 following twice daily dosing, with a mean (% coefficient of variation [%CV]) accumulation ratio of 1.5 (41%).

Absorption

After administration of RAHIKA 400 mg orally in patients with cancer, capmatinib peak plasma concentrations (C_{max}) were reached in approximately 1 to 2 hours (T_{max}). The absorption of capmatinib after oral administration is estimated to be greater than 70%.

Effect of Food

A high-fat meal (containing approximately 1000 calories and 50% fat) in healthy subjects increased capmatinib AUC_{0-INF} by 46% with no change in C_{max} compared to under fasted conditions. A low-fat meal (containing approximately 300 calories and 20% fat) in healthy subjects had no clinically meaningful effect on capmatinib exposure. When capmatinib was administered at 400 mg orally twice daily in cancer patients, exposure (AUC_{0-12h}) was similar after administration of capmatinib with food and under fasted conditions.

Distribution

Capmatinib plasma protein binding is 96%, independent of capmatinib concentration. The apparent mean volume of distribution at steady-state is 164 L.

The blood-to-plasma ratio was 1.5, but decreased at higher concentrations to 0.9.

Elimination

The effective elimination half-life of capmatinib is 6.5 hours. The mean (%CV) steady-state apparent clearance of capmatinib is 24 L/hr (82%).

Metabolism

Capmatinib is primarily metabolized by CYP3A4 and aldehyde oxidase.

Excretion

Following a single oral administration of radiolabeled-capmatinib to healthy subjects, 78% of the total radioactivity was recovered in feces with 42% as unchanged and 22% was recovered in urine with negligible as unchanged.

Specific Populations

No clinically significant effects on the pharmacokinetic parameters of capmatinib were identified for the following covariates assessed: age (26 to 90 years), sex, race (White, Asian, Native American, Black, unknown), body weight (35 to 131 kg), mild to moderate renal impairment (baseline CLcr 30 to 89 mL/min by Cockcroft-Gault) and mild, moderate or severe hepatic impairment (Child-Pugh classification). The effect of severe renal impairment (baseline CLcr 15 to 29 mL/min) on capmatinib pharmacokinetics has not been studied.

Drug Interaction Studies

Clinical Studies and Model-Informed Approaches

Strong CYP3A Inhibitors: Coadministration with itraconazole (a strong CYP3A inhibitor) increased capmatinib AUC_{0-INF} by 42% with no change in capmatinib C_{max} .

Strong CYP3A Inducers: Coadministration with rifampicin (a strong CYP3A inducer) decreased capmatinib AUC_{0-INF} by 67% and decreased C_{max} by 56%.

Moderate CYP3A Inducers: Coadministration with efavirenz (a moderate CYP3A inducer) was predicted to decrease capmatinib AUC_{0-12h} by 44% and decrease C_{max} by 34%.

Proton Pump Inhibitors: Coadministration with rabeprazole (a proton pump inhibitor) decreased capmatinib AUC_{0-INF} by 25% and decreased C_{max} by 38%.

Substrates of CYP Enzymes: Coadministration of capmatinib increased caffeine (a CYP1A2 substrate) AUC_{0-INF} by 134% with no change in its C_{max}. Coadministration of capmatinib had no clinically meaningful effect on exposure of midazolam (a CYP3A substrate).

P-gp Substrates: Coadministration of capmatinib increased digoxin (a P-gp substrate) AUC_{0-INF} by 47% and increased C_{max} by 74%.

BCRP Substrates: Coadministration of capmatinib increased rosuvastatin (a BCRP substrate) AUC_{0-INF} by 108% and increased C_{max} by 204%.

In Vitro Studies

Transporter Systems: Capmatinib is a substrate of P-gp, but not a substrate of BCRP or MRP2. Capmatinib reversibly inhibits MATE1 and MATE2K, but does not inhibit OATP1B1, OATP1B3, OCT1, OAT1, OAT3, or MRP2.

13 NONCLINICAL TOXICOLOGY

13.1 Carcinogenesis, Mutagenesis, Impairment of Fertility

Carcinogenicity studies were not conducted with capmatinib. Capmatinib was not mutagenic in an in vitro bacterial reverse mutation assay and did not cause chromosomal aberrations in an in vitro chromosome aberration assay in human peripheral blood lymphocytes. Capmatinib was not clastogenic in an in vivo bone marrow micronucleus test in rats.

Dedicated fertility studies were not conducted with capmatinib. No effects on male and female reproductive organs occurred in general toxicology studies conducted in rats and monkeys at doses resulting in exposures of up to approximately 3.6 times the human exposure based on AUC at the 400 mg twice daily clinical dose.

13.2 Animal Toxicology and/or Pharmacology

In rats, capmatinib administration resulted in vacuolation of white matter of the brain in both 4- and 13-week studies at doses ≥ 2.2 times the human exposure (AUC) at the 400 mg twice daily clinical dose. In some cases, the brain lesions were associated with early death and/or convulsions or tremors. Concentrations of capmatinib in the brain tissue of rats was approximately 9% of the corresponding concentrations in plasma.

In vitro and in vivo assays demonstrated that capmatinib has some potential for photosensitization; however, the no-observed-adverse-effect level for in vivo photosensitization was 30 mg/kg/day (C_{max} of 14000 ng/mL), about 2.9 times the human C_{max} at the 400 mg twice daily clinical dose.

14 CLINICAL STUDIES

Metastatic NSCLC with a Mutation that Leads to MET Exon 14 Skipping

The efficacy of RAHIKA was evaluated in GEOMETRY mono-1, a multicenter, non-randomized, open-label, multi-cohort study (NCT02414139). Eligible patients were required to have NSCLC with a mutation that leads to MET exon 14 skipping, epidermal growth factor receptor (EGFR) wild-type and anaplastic lymphoma kinase (ALK) negative status, and at least one measurable lesion as defined by Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1. Patients with symptomatic CNS metastases, clinically significant uncontrolled cardiac disease, or who received treatment with any MET or hepatocyte growth factor (HGF) inhibitor were not eligible for the study.

Out of the first 97 patients enrolled in GEOMETRY mono-1 following the central confirmation of MET exon 14 skipping by a RNA-based clinical trial assay, 78 patient samples were retested with the FDA-approved FoundationOne® CDx (22 treatment-naïve and 56 previously treated patients) to detect mutations that lead to MET exon 14 skipping. Out of 78 samples retested with FoundationOne® CDx, 73 samples were evaluable (20 treatment-naïve and 53 previously treated patients), 72 (20 treatment-naïve and 52 previously treated patients) of which were confirmed to have a mutation that leads to MET exon 14 skipping, demonstrating an estimated positive percentage agreement of 99% (72/73) between the clinical trial assay and the FDA-approved assay.

Patients received RAHIKA 400 mg orally twice daily until disease progression or unacceptable toxicity. The major efficacy outcome measure was overall response rate (ORR) as determined by a Blinded Independent Review Committee (BIRC) according to RECIST 1.1. An additional efficacy outcome measure was duration of response (DOR) by BIRC.

The efficacy population included 60 treatment-naïve patients and 100 previously treated patients. The median age was 71 years (range: 48 to 90 years); 61% female; 77% White; 25% had Eastern Cooperative Oncology Group (ECOG) Performance Status (PS) 0 and 74% had ECOG PS 1; 61% never smoked; 83% had adenocarcinoma; and 16% had CNS metastases. Among previously treated patients, 81% received one, 16% received two and 3% received three prior lines of systemic therapy. Amongst previously treated patients, 86% received prior platinum-based chemotherapy.

Efficacy results are presented in Table 5.

Table 5: Efficacy Results for Treatment-Naïve and Previously Treated Patients in GEOMETRY mono-1

Efficacy parameters	Treatment-Naïve N = 60	Previously treated N = 100
Overall response rate^{a,b} (95% CI)^c	68% (55, 80)	44% (34, 54)
Complete response	5%	0
Partial response	63%	44%
Duration of response (DOR)^a		
Median (months) (95% CI) ^d	16.6 (8.4, 22.1)	9.7 (5.6, 13.0)
Patients % with DOR $\geq$ 12 months	49%	36%

Abbreviations: CI, confidence interval; NE, not estimable.

^aBlinded Independent Review Committee (BIRC) review.

^bConfirmed response.

^cClopper and Pearson exact binomial 95% CI.

^dBased on Kaplan-Meier estimate.

STORAGE

Do not store above 30°C. Protect from moisture.

RAHIKA should not be used after the date marked “EXP” on the pack.

RAHIKA must be kept out of the reach and sight of children.

PACK SIZE

Available in PVC/PCTFE blister film backed by Aluminium foil as backing component in pack of 60 tablets.

PRODUCT REGISTRATION HOLDER:

Novartis Corporation (M) Sdn. Bhd.
Level 18, Imazium,
No. 8, Jalan SS21/37,
Damansara Uptown,
47400 Petaling Jaya,
Selangor.

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